Pet Care Ecosystem Platform

Technical Scope, Architecture & Operational Proposal

Executive Overview

This proposal outlines the complete architecture, development scope, operational structure, and deployment strategy for a modern pet care ecosystem platform designed to support long-term scalability, operational efficiency, and future business expansion.

The platform is planned as a complete digital ecosystem that combines pet service bookings, ecommerce management, real-time operations, adoption services, lost & found reporting, customer engagement, and dynamic content management under one unified infrastructure.

The entire system is being designed with production-grade architecture standards to ensure the platform remains scalable, maintainable, secure, and migration-ready for future cloud infrastructure expansion.

The platform will consist of:

- **Customer Mobile Application**
- **Service Provider/Vendor Mobile Application**
- **Centralized Admin & CMS Panel**
- **Realtime Communication Infrastructure**
- **Ecommerce & Inventory Management**
- **Adoption & Lost/Found Management**
- **Booking & Operational Systems**
- **Wallet, Payments & Revenue Management**

Technology Stack

Mobile Applications

The customer and provider applications will be developed using Flutter, allowing a single scalable codebase for both Android and iOS platforms while maintaining native-level performance and smooth UI responsiveness.

Mobile Applications Included

1. **Customer/User Application**
2. **Service Provider/Vendor Application**

Mobile Technology

- **Flutter**
- **Firebase Cloud Messaging (FCM)**
- **Socket.io for realtime functionality**
- **Google Maps Integration**
- **API-driven architecture**

The Flutter architecture will be modular to support future feature additions without affecting existing functionality.

Backend Infrastructure

The primary backend system will be developed using Node.js with NestJS.

This architecture has been selected specifically for:

- **High scalability**
- **Enterprise-grade modularity**
- **Strong API structuring**
- **Realtime handling capability**
- **Future microservice readiness**
- **Long-term maintainability**

Core Backend Responsibilities

The backend infrastructure will manage:

- **Authentication & authorization**
- **Mobile APIs**
- **Booking systems**
- **Ecommerce operations**
- **Payment processing**
- **Wallet management**
- **Pet profile management**
- **Vendor management**
- **Notifications**
- **Realtime booking updates**
- **Chat system integrations**
- **Adoption management**
- **Lost & found services**

The backend will follow a clean modular architecture with reusable service layers and isolated business logic.

Realtime Infrastructure

Realtime operations will be handled separately using Socket.io.

This allows optimized event-based communication without overloading the main API layer.

Realtime Features

- **Live booking updates**
- **Live provider tracking**
- **Customer/provider chat**
- **Booking status changes**
- **Instant notifications**
- **Delivery/service progress updates**
- **Live operational events**

Redis will support realtime event broadcasting and queue handling for better performance.

Admin Panel & Operational CMS

The administrative and operational management panel will be developed using Laravel.

Laravel is being selected specifically for its flexibility in CMS management, operational dashboards, RBAC implementation, and dynamic admin workflows.

The admin panel is designed to allow complete operational management without requiring daily developer involvement.

This means the management team can independently:

- **Manage homepage content**
- **Update banners**
- **Control products**
- **Handle vendors**
- **Manage bookings**
- **Configure offers**
- **Control notifications**
- **Adjust operational settings**
- **Generate reports**

The admin panel will not handle realtime processing or mobile application APIs directly.

Its purpose is focused on:

- **Administrative operations**
- **CMS management**
- **Business analytics**
- **User management**
- **Operational monitoring**

Database Architecture

PostgreSQL

PostgreSQL will serve as the primary database.

It will manage:

- **Users**
- **Vendors**
- **Pets**
- **Bookings**
- **Ecommerce orders**
- **Transactions**
- **Adoption records**
- **Lost/found records**

- **Reviews**
- **CMS data**
- **Reports**

The schema structure will be optimized for relational scalability and future analytics expansion.

Redis

Redis will be used as a high-performance caching and queue layer.

Redis responsibilities include:

- **OTP storage**
- **Queue processing**
- **Session caching**
- **Realtime data handling**
- **Notification queues**
- **Temporary data storage**
- **Performance optimization**

This setup significantly improves system responsiveness under high traffic.

File Storage Infrastructure

Cloudflare R2 will be used for file and media storage.

The storage layer will remain abstracted so the platform can migrate to AWS S3 in the future without major code restructuring.

Stored Assets

- **User images**
- **Pet images**
- **Product images**
- **Documents**
- **Vendor verification files**
- **Banner media**
- **Promotional content**
- **Videos**

This approach ensures scalable and cost-efficient media handling.

Server & Deployment Architecture

The infrastructure is designed to support both immediate VPS deployment and future enterprise cloud migration.

Initial Infrastructure

- **VPS Deployment**
- **Nginx Web Server**
- **Dockerized Services**
- **PM2 Process Management**
- **SSL Security Layer**
- **CI/CD Ready Environment**

Future Migration Readiness

The architecture is intentionally designed for future migration toward:

- **AWS ECS/EKS**
- **AWS RDS**
- **AWS ElastiCache**
- **AWS S3**
- **Load-balanced infrastructure**
- **CDN distribution**
- **Horizontal scaling**

No major codebase restructuring will be required during future infrastructure migration.

PETSEWA

Enterprise Platform Scope of Work & Architecture Overview

Project Vision

PetSewa is being designed as a complete hyperlocal pet ecosystem that connects pet owners, veterinarians, groomers, trainers, hostels, dog walkers, pet stores, shelters, NGOs, content creators, and community members through a unified digital platform.

The platform will consist of:

- **Customer Mobile Application**
- **Vendor Mobile Application**
- **Full Web Platform**
- **Ecommerce Marketplace**
- **Hyperlocal Service Marketplace**
- **Social Media Platform**
- **Reels Platform**
- **Community Platform**
- **NGO & Fundraising Platform**
- **Adoption Platform**
- **Lost & Found Platform**
- **Emergency Veterinary Connect System**
- **Rewards & Referral Ecosystem**
- **Advertisement & Promotion Marketplace**
- **Centralized Admin Panel**
- **Vendor Management Portal**

The architecture is designed to be modular, scalable, and future-ready for nationwide expansion.

Platform Components

Customer Application

The customer application will serve as the primary interface for pet owners.

Core Modules

- **User Registration & Authentication**
- **Pet Profile Management**
- **Hyperlocal Service Discovery**
- **Booking Management**
- **Ecommerce Shopping**
- **Adoption Portal**
- **Lost & Found Portal**
- **NGO Donations**
- **Social Feed**
- **Reels Platform**
- **Community Groups**
- **Referral & Rewards**
- **Wallet System**
- **Emergency Veterinary Connect**

Vendor Application

A single role-based vendor application will support:

- **Veterinarians**
- **Groomers**
- **Trainers**
- **Boarding Providers**
- **Hostels**
- **Dog Walkers**
- **Pet Stores**
- **Shelters**

Vendor Features

- **Booking Management**
- **Service Management**
- **Availability Management**
- **Revenue Dashboard**
- **Wallet Management**
- **Payout Requests**
- **Reviews & Ratings**
- **Staff Management**
- **Customer History**
- **Business Profile Management**

Specialized Modules

Veterinary Module

- **Pet Medical Records**
- **Vaccination Records**
- **Prescriptions**
- **Treatment Notes**
- **Inventory Management**

Groomer Module

- **Grooming Packages**
- **Before/After Gallery**
- **Grooming History**

Trainer Module

- **Training Sessions**
- **Progress Reports**

- **Behaviour Tracking**

Boarding & Hostel Module

- **Occupancy Tracking**
- **Room Management**
- **Check-In / Check-Out**
- **Medication Logs**

Dog Walker Module

- **Walk Scheduling**
- **Walk Tracking**
- **Daily Reports**

Hyperlocal Location Engine

The entire PetSewa ecosystem will operate on a location-first architecture.

First-Time User Flow

Upon first launch:

- **Allow GPS Location**
- **Enter City**
- **Enter Area**
- **Enter Pincode**

Location selection is mandatory before accessing services.

Stored Data

- **Latitude**
- **Longitude**
- **City**
- **Area**
- **Pincode**

Hyperlocal Features

- **Nearby Veterinarians**
- **Nearby Groomers**
- **Nearby Trainers**
- **Nearby Hostels**
- **Nearby NGOs**

- **Nearby Pet Stores**
- **Nearby Adoption Listings**
- **Nearby Lost & Found Cases**
- **Nearby Emergency Services**

Search Capabilities

- **Radius Search**
- **Distance Search**
- **Geo-Fencing Ready**
- **Location-Based Feed Personalization**

Emergency Prioritization

Emergency veterinary requests automatically prioritize:

- **Closest Doctors**
- **Available Doctors**
- **Emergency-Certified Doctors**

Website Ecosystem Platform

The website will function as a complete PetSewa ecosystem and not merely a promotional website.

Homepage

- **Hero Slider**
- **Service Discovery**
- **Featured Veterinarians**
- **Featured Groomers**
- **Featured Trainers**
- **Featured Hostels**
- **Featured Products**
- **Adoption Highlights**
- **Lost & Found Highlights**
- **NGO Campaigns**
- **Community Feed**
- **Trending Reels**
- **Featured Creators**
- **Blogs**
- **Success Stories**
- **Download App CTA**

Additional Website Modules

- **Services Portal**
- **Vendor Directory**
- **Ecommerce Marketplace**
- **Adoption Platform**
- **Lost & Found Platform**
- **NGO Fundraising Platform**
- **Social Media Platform**
- **Community Platform**
- **Blog & Knowledge Center**
- **Careers**
- **Partner With Us**
- **Advertise With Us**

Ecommerce Marketplace

Customer Features

- **Product Catalog**
- **Categories & Subcategories**
- **Wishlist**
- **Cart**
- **Checkout**
- **Coupons**
- **Order Tracking**
- **Reviews**
- **Refund Requests**

Vendor Features

- **Product Management**
- **Inventory Management**
- **Order Fulfillment**
- **Sales Reports**

Hyperlocal Commerce

- **City-Based Inventory**
- **Delivery Area Management**
- **Seller Prioritization**
- **Delivery Estimation**

Adoption Platform

Features

- **Adoption Listings**
- **Pet Profiles**
- **Application Management**
- **Shelter Profiles**
- **Adoption Success Stories**

Pet Profiles

- **Photos**
- **Videos**
- **Breed**
- **Age**
- **Gender**
- **Vaccination Status**
- **Health Information**

Lost & Found Platform

Features

- **Lost Pet Reports**
- **Found Pet Reports**
- **Location Mapping**
- **Image Uploads**
- **Contact Management**

Future Expansion

- **AI Image Matching**
- **Location-Based Matching Engine**

NGO & Fundraising Platform

NGO Management

- **NGO Registration**

- **Document Verification**
- **Approval Workflow**
- **NGO Profiles**

Fundraising

- **Campaign Creation**
- **Donation Collection**
- **Progress Tracking**
- **Donor Reports**
- **Fund Utilization Reports**

Donations

- **UPI**
- **Cards**
- **Net Banking**

Social Media Platform

Features

- **User Profiles**
- **Pet Profiles**
- **Posts**
- **Likes**
- **Comments**
- **Shares**
- **Saved Posts**
- **Hashtags**
- **Mentions**

Feed Types

- **Following Feed**
- **Trending Feed**
- **Recommended Feed**
- **Nearby Feed**

Reels Platform

Creator Tools

- **Video Upload**
- **Recording**
- **Trimming**
- **Editing**
- **Thumbnail Selection**

Engagement

- **Likes**
- **Comments**
- **Shares**
- **Saves**
- **Follow Creator**

Community Platform

Community Types

- **Dog Communities**
- **Cat Communities**
- **Breed Communities**
- **Rescue Communities**
- **Adoption Communities**

Features

- **Discussions**
- **Events**
- **Polls**
- **Announcements**

Rewards, Referral & Gamification

Reward Activities

- **Bookings**
- **Purchases**
- **Referrals**
- **Social Activity**

- **Adoption Participation**

Features

- **Pet Points**
- **User Levels**
- **Achievement Badges**
- **Daily Check-In Rewards**
- **Leaderboards**

Levels

- **Bronze**
- **Silver**
- **Gold**
- **Platinum**

Advertisement & Promotion Marketplace

Promotion Types

- **Featured Listings**
- **Sponsored Listings**
- **Homepage Placements**
- **Sponsored Products**
- **Sponsored Posts**
- **Sponsored Reels**

Vendor Dashboard

- **Campaign Management**
- **Click Analytics**
- **Impression Analytics**
- **Revenue Tracking**

Revenue Models

- **Featured Plans**
- **Sponsored Plans**
- **Banner Advertisements**
- **Homepage Placements**

Emergency Veterinary Connect

User Features

- **SOS Button**
- **Emergency Categories**
- **Emergency Vet Search**
- **Emergency Chat**
- **Emergency Payments**

Doctor Features

- **Emergency Duty Toggle**
- **Emergency Case Handling**
- **Emergency Pricing**
- **Medical Record Access**

Admin Features

- **Emergency Vet Approval**
- **Scheduling**
- **Monitoring Dashboard**
- **Performance Tracking**

Financial & Commission Engine

Payment Methods

- **UPI**
- **Debit/Credit Cards**
- **Net Banking**
- **Wallet**

Commission System

Fully Dynamic Category-Based Commission Structure

Examples:

- **Veterinarian**
- **Grooming**

- **Training**
- **Boarding**
- **Dog Walking**
- **Ecommerce Products**

Commission rules configurable through Admin Panel.

Vendor Wallet

- **Earnings**
- **Pending Amount**
- **Withdrawals**
- **Payout Requests**
- **Transaction History**

Enterprise Admin Panel

Core Modules

- **Dashboard**
- **User Management**
- **Vendor Management**
- **Booking Management**
- **Ecommerce Management**
- **Adoption Management**
- **Lost & Found Management**
- **NGO Management**
- **Fundraising Management**
- **Community Moderation**
- **Social Moderation**
- **Reels Moderation**
- **Rewards Management**
- **Advertisement Management**
- **Finance Management**
- **CMS Management**
- **SEO Management**
- **Analytics & Reporting**

Enterprise Architecture

Frontend Layer

Mobile Applications

- **Flutter**

Web Platform

- **Next.js**

Vendor Portal

- **Next.js**

Admin Portal

- **Laravel**

Backend Layer

NestJS

Responsible For:

- **Authentication**
- **Booking Engine**
- **Ecommerce Engine**
- **Social Media Engine**
- **Reels Engine**
- **Community Engine**
- **Donation Engine**
- **Emergency Module**
- **Wallet System**
- **Rewards Engine**
- **Notification Engine**

Realtime Infrastructure

Socket.io

Used For:

- **Chat**
- **Emergency Requests**
- **Booking Updates**
- **Feed Updates**
- **Notifications**

Redis Pub/Sub will support realtime scalability.

Database Layer

PostgreSQL

Core Databases:

- **Users**
- **Vendors**
- **Bookings**
- **Products**
- **Orders**
- **Pets**
- **Medical Records**
- **Donations**
- **Posts**
- **Reels**
- **Communities**
- **Rewards**
- **Promotions**
- **Wallets**

Performance Layer

Redis

Used For:

- **Cache**
- **OTP Storage**
- **Session Management**
- **Feed Caching**
- **Geo Queries**
- **Queue Processing**

Storage Layer

Cloudflare R2

Used For:

- **Images**

- **Videos**
- **Reels**
- **Documents**
- **Medical Records**
- **Campaign Media**

Storage abstraction maintained for future AWS S3 migration.

Infrastructure

Nginx
PM2
Docker
VPS Deployment

Future Ready:

- **AWS**
- **Kubernetes**
- **Load Balancing**
- **CDN Scaling**
- **Multi-Region Deployment**

Scalability Requirements

The architecture will support:

- **Multi-Vendor Expansion**
- **Dynamic Vendor Categories**
- **Dynamic Commission Rules**
- **RBAC Permissions**
- **Multi-City Expansion**
- **Hyperlocal Search**
- **Geo-Fencing**
- **AI Recommendations**
- **Creator Monetization**
- **Live Streaming**
- **Subscription Plans**
- **Sponsored Advertising Ecosystem**

All modules will be fully configurable through the Admin Panel without requiring regular developer intervention.

GitHub & Development Workflow

The project will follow structured version control practices.

Branch Structure

- **Main**
- **Development**
- **Staging**
- **Feature branches**
- **Hotfix branches**

Development Practices

- **Code reviews**
- **Modular architecture**
- **CI/CD readiness**
- **Environment-based deployment**
- **Documentation standards**

Deliverables

The following deliverables will be included within the project scope.

Technical Deliverables

- **Complete source code**
- **Mobile applications**
- **Backend APIs**
- **Laravel admin panel**
- **Database architecture**
- **Deployment configurations**
- **Docker setup**
- **VPS deployment support**

Documentation Deliverables

- **Architecture documentation**

- **API documentation**
- **Database schema documentation**
- **Deployment documentation**
- **Admin panel usage documentation**

Design Deliverables

- **Admin wireframes**
- **Application flow structure**
- **CMS management flow**

Estimated Development Timeline

Phase 1 — Planning & Architecture

Estimated Duration: 1 Week

Includes:

- **Requirement analysis**
- **Technical planning**
- **Database planning**
- **Infrastructure planning**

Phase 2 — UI/UX Design

Estimated Duration: 2 Weeks

Includes:

- **Mobile UI design**
- **Admin dashboard wireframes**
- **User experience mapping**

Phase 3 — Backend Development

Estimated Duration: 5–7 Weeks

Includes:

- **API development**
- **Authentication**
- **Booking systems**

- **Ecommerce systems**
- **Wallet systems**
- **Realtime integrations**

Phase 4 — Mobile Application Development

Estimated Duration: 6–8 Weeks

Includes:

- **Flutter app development**
- **API integration**
- **Realtime implementation**
- **Notifications**

Phase 5 — Admin Panel Development

Estimated Duration: 4–5 Weeks

Includes:

- **Dashboard systems**
- **CMS infrastructure**
- **Analytics systems**
- **Operational controls**

Phase 6 — Additional Modules

Estimated Duration: 2–3 Weeks

Includes:

- **Lost & found module**
- **Adoption module**
- **Final operational workflows**

Phase 7 — QA, Testing & Deployment

Estimated Duration: 2 Weeks

Includes:

- **Testing**
- **Bug fixing**
- **Performance optimization**
- **Production deployment**

Estimated Total Timeline

Approximately 2.5-3 months depending on final feature approval and revisions.

Recommended Team Structure

- **Project Manager**
- **UI/UX Designer**
- **Flutter Developers**
- **Backend Developers**
- **Laravel Developer**
- **QA Engineer**
- **DevOps Engineer**

Long-Term Scalability Vision

The platform is intentionally being built as a scalable ecosystem rather than a basic application.

The infrastructure supports:

- **Future AI integrations**
- **Subscription systems**
- **Franchise/vendor expansion**
- **Multi-city operations**
- **AWS migration**
- **Advanced analytics**
- **IoT/GPS integrations**
- **Loyalty & referral systems**
- **Multi-language support**

This ensures the platform remains technically sustainable as the business grows.

Final Scope Summary

The proposed system includes:

- **Customer mobile application**
- **Provider mobile application**
- **Enterprise-grade backend infrastructure**
- **Realtime communication systems**
- **Dynamic Laravel admin CMS**
- **Ecommerce ecosystem**
- **Wallet & payment systems**
- **Lost & found module**
- **Adoption platform**
- **Reporting & analytics**
- **Vendor management**
- **Deployment architecture**
- **Full documentation**
- **Scalable production-ready infrastructure**

The overall architecture is designed to support operational independence, future scalability, cloud migration readiness, and long-term maintainability while minimizing future redevelopment requirements.

Website & Ecommerce Platform Scope

Website Purpose & Business Objective

The website will function as a high-performance marketing, ecommerce, and conversion platform designed to:

- **Present the complete ecosystem professionally**
- **Showcase platform services & features**
- **Drive mobile app downloads**

- **Support ecommerce sales**
- **Improve SEO visibility**
- **Build trust & brand authority**
- **Support customer acquisition campaigns**
- **Provide web-based shopping functionality**

The platform strategy is designed to keep high-engagement community and realtime interactions primarily inside the mobile applications while allowing the website to act as the primary acquisition and commerce channel.

Website Structure

1. Landing Website

A premium responsive website will be developed to professionally showcase the platform ecosystem.

Main Sections

- **Hero section with app download CTA**
- **Platform overview**
- **Service categories**
- **Pet care solutions**
- **Adoption system showcase**
- **Lost & found feature showcase**
- **Social community feature highlights**
- **Ecommerce highlights**
- **Vendor/partner onboarding**
- **Testimonials & reviews**
- **Blog/news section**
- **Contact & support**
- **App Store / Play Store integrations**

The landing website will be fully responsive and optimized for mobile, tablet, and desktop devices.

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2. Full Ecommerce Website

The web platform will include complete ecommerce functionality connected directly with the backend infrastructure.

Ecommerce Features

- **Product catalog**
- **Categories & subcategories**
- **Product search & filters**
- **Wishlist**
- **Cart & checkout**
- **Secure payments**
- **Coupons & promotions**
- **Order tracking**
- **Product reviews & ratings**
- **Featured products**
- **Dynamic offers**
- **Inventory management**
- **SEO-friendly product pages**

The ecommerce system will synchronize with:

- **Mobile applications**
- **Admin panel**
- **Vendor dashboards**
- **Inventory systems**

in realtime.

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3. Social Media & Community Integration

The website will showcase the social ecosystem while encouraging users to continue engagement through the mobile applications.

Web Social Features

- **Public pet community feed**
- **Featured posts**
- **Community highlights**
- **Creator/influencer showcase**
- **Trending pet content**
- **Adoption stories**
- **Educational pet content**
- **Shareable social pages**

App-Focused Features

High-engagement social features such as:

- **Realtime chat**
- **Reels**
- **Infinite feed scrolling**
- **Live interactions**
- **Push notifications**
- **Community engagement**

will primarily remain app-focused to improve retention and daily active usage.

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4. Lost & Found Web Portal

The website will provide public accessibility for lost & found reports.

Features

- **Lost pet listings**
- **Found pet listings**
- **Search & filters**
- **Location-based discovery**
- **Report submission**
- **Public sharing support**
- **Community visibility**

This improves public reach and search engine visibility.

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5. Pet Adoption Portal

A dedicated web adoption section will allow users to:

- **Browse adoption listings**
- **View pet profiles**
- **Submit adoption requests**
- **Contact shelters/providers**
- **View pet details & health information**

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6. Vendor & Partner Landing Sections

Dedicated sections/pages for:

- **Groomers**
- **Trainers**
- **Veterinary partners**
- **Shelters**
- **Pet stores**
- **Service providers**

with onboarding forms and partnership applications.

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7. SEO & Marketing Infrastructure

The website will be built with SEO-focused architecture using Next.js for maximum visibility and performance.

SEO Features

- **Fast-loading pages**
- **SEO-friendly URLs**
- **Dynamic metadata**
- **Structured schema markup**
- **Sitemap generation**
- **Blog optimization**
- **Open Graph integration**
- **Google indexing optimization**

This helps improve:

- **Organic traffic**
- **Google rankings**
- **Brand discoverability**
- **Customer acquisition**

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8. Recommended Website Technology Stack

Frontend

- **Next.js (React)**

Backend Integration

- **NestJS APIs**
- **Socket.io**
- **PostgreSQL**
- **Redis**
- **Cloudflare R2**

Deployment

- **Docker**
- **Nginx**
- **VPS/AWS-ready infrastructure**

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9. Business Strategy Behind the Architecture

The architecture is intentionally designed to:

Website

Focus on:

- **Brand visibility**
- **SEO**
- **Ecommerce**
- **Marketing**
- **User acquisition**
- **Public discoverability**

Mobile Apps

Focus on:

- **Daily engagement**
- **Realtime interactions**
- **Social activity**
- **Booking management**
- **Notifications**
- **Community retention**

This creates a scalable ecosystem optimized for both acquisition and long-term engagement.

Final Website Deliverables

The website scope will include:

- **Premium landing website**
- **Full ecommerce store**
- **SEO optimization**
- **CMS-based content management**
- **Public social/community sections**
- **Adoption portal**
- **Lost & found portal**
- **Vendor onboarding pages**
- **Blog/news system**
- **Mobile app download integrations**
- **Dynamic homepage management**
- **Analytics integration**
- **Marketing-ready architecture**

The entire platform will remain fully scalable and synchronized with the mobile apps, backend systems, and admin infrastructure.

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Technical Architecture View

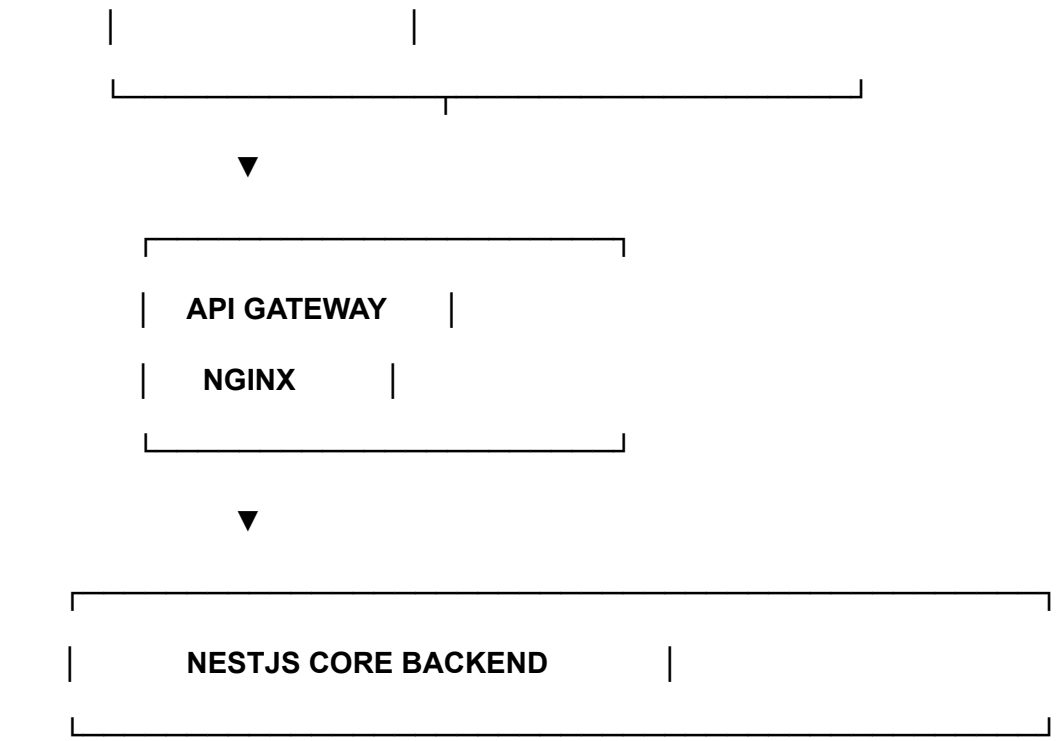
High-Level System Architecture

PETSEWA ECOSYSTEM

CUSTOMER CHANNELS

Flutter App **Website (Next.js)**

(iOS / Android) **(Full Ecosystem Platform)**



Authentication Service **Booking Engine** **Ecommerce Engine** **Social Media Engine**

Business Service Layer

- **Vendor Management**

- **Adoption System**
- **Lost & Found System**
- **NGO Management**
- **Fundraising System**
- **Rewards & Referral Engine**
- **Wallet System**
- **Commission Engine**
- **Advertisement Engine**
- **Emergency Vet Connect**
- **Notification Engine**
- **CMS APIs**
- **Search Engine**

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SOCKET.IO REALTIME HUB

- **Chat**
- **Emergency Requests**
- **Booking Updates**
- **Notifications**
- **Realtime Feed Updates**

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PostgreSQL **Redis** **Cloudflare R2** **Firebase**

Database **Cache** **Media Storage** **Notifications**

Hyperlocal Location Architecture

Location is the foundation of the entire platform.

User Opens App / Website

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Location Permission Popup

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│ Allow GPS Location │

│ Enter City/Area/Pincode │

│ Not Now │

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Location Processing

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│ GPS Coordinates │

│ Latitude │

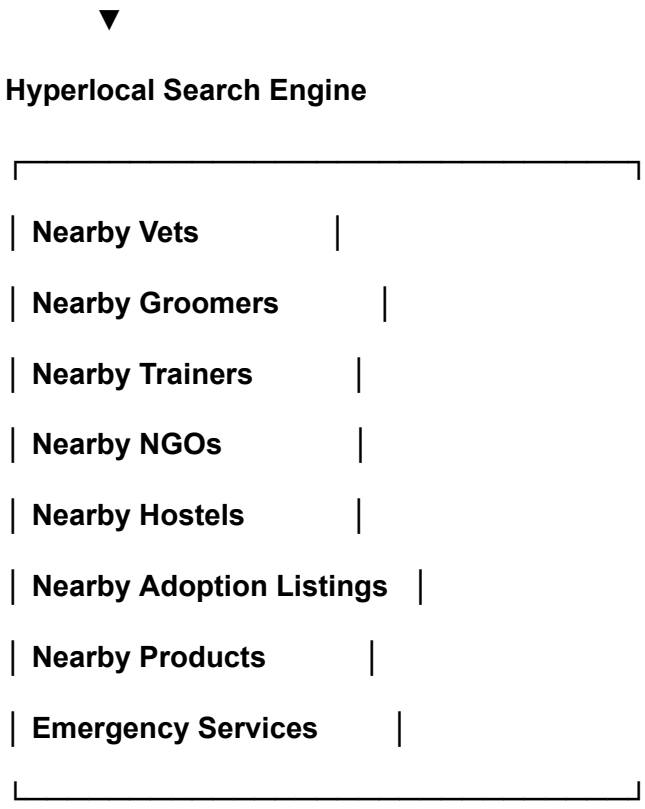
│ Longitude │

│ City │

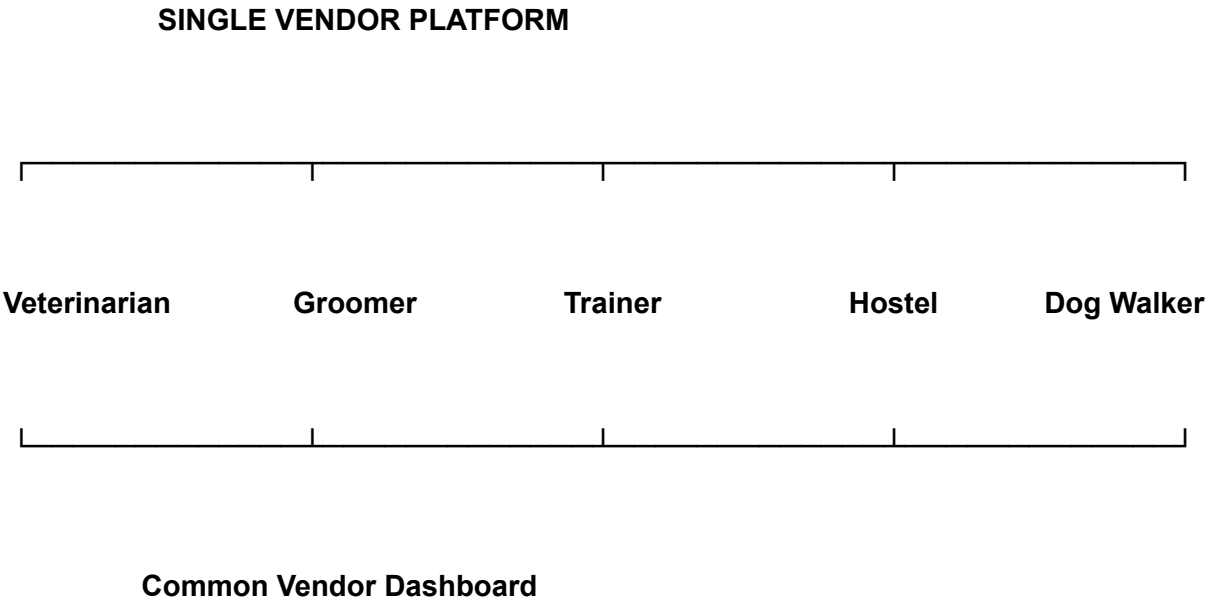
│ Area │

│ Pincode │

└──────────────────────────────────────┘



Vendor Ecosystem Architecture



• Bookings

• Earnings

• Wallet

• Reports

• Customers

• Reviews

• Staff Management

▼

Role-Based Permissions

▼

Specialized Modules

Doctor → Medical Records

Groomer → Grooming Packages

Trainer → Training Sessions

Hostel → Occupancy Management

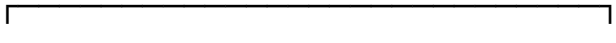
Walker → Walk Scheduling

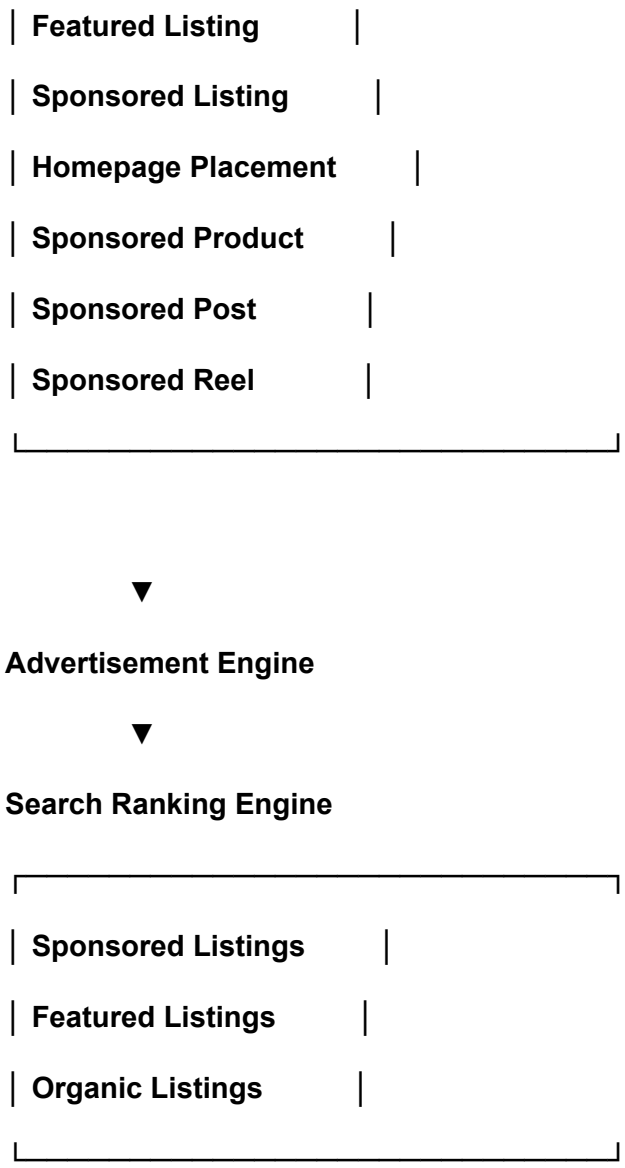
Advertisement & Revenue Architecture

Vendor / Doctor / Store Owner

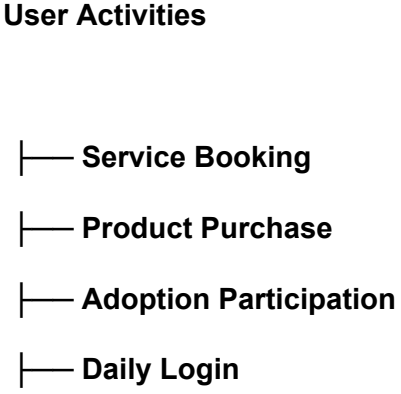
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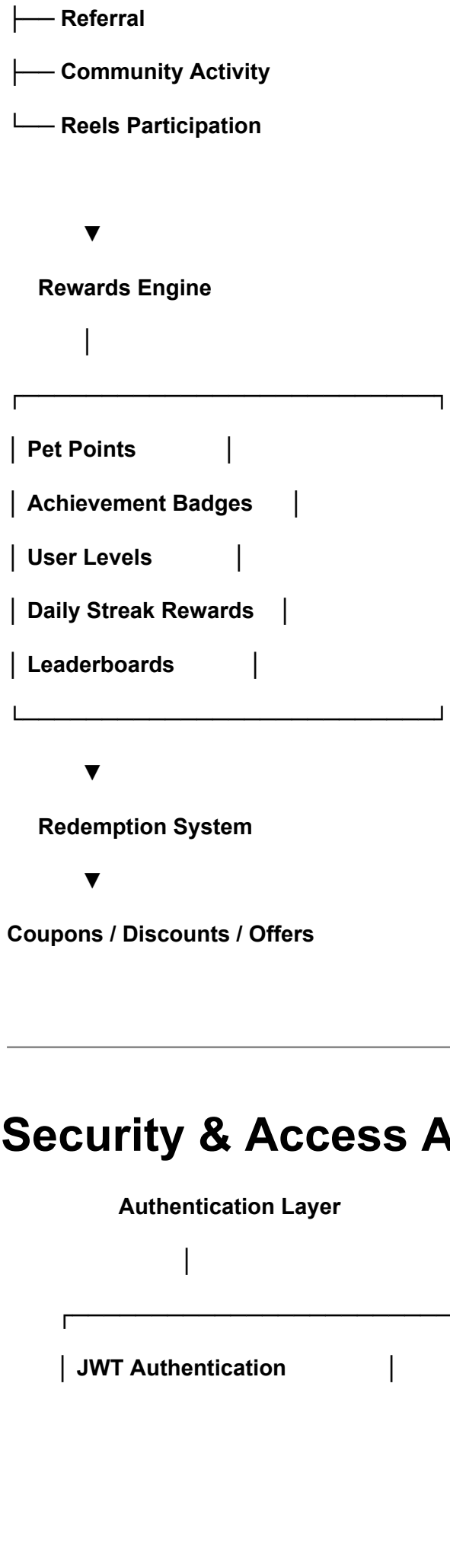
Promotion Marketplace



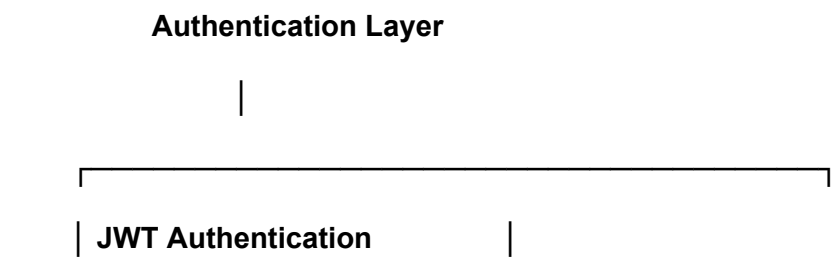


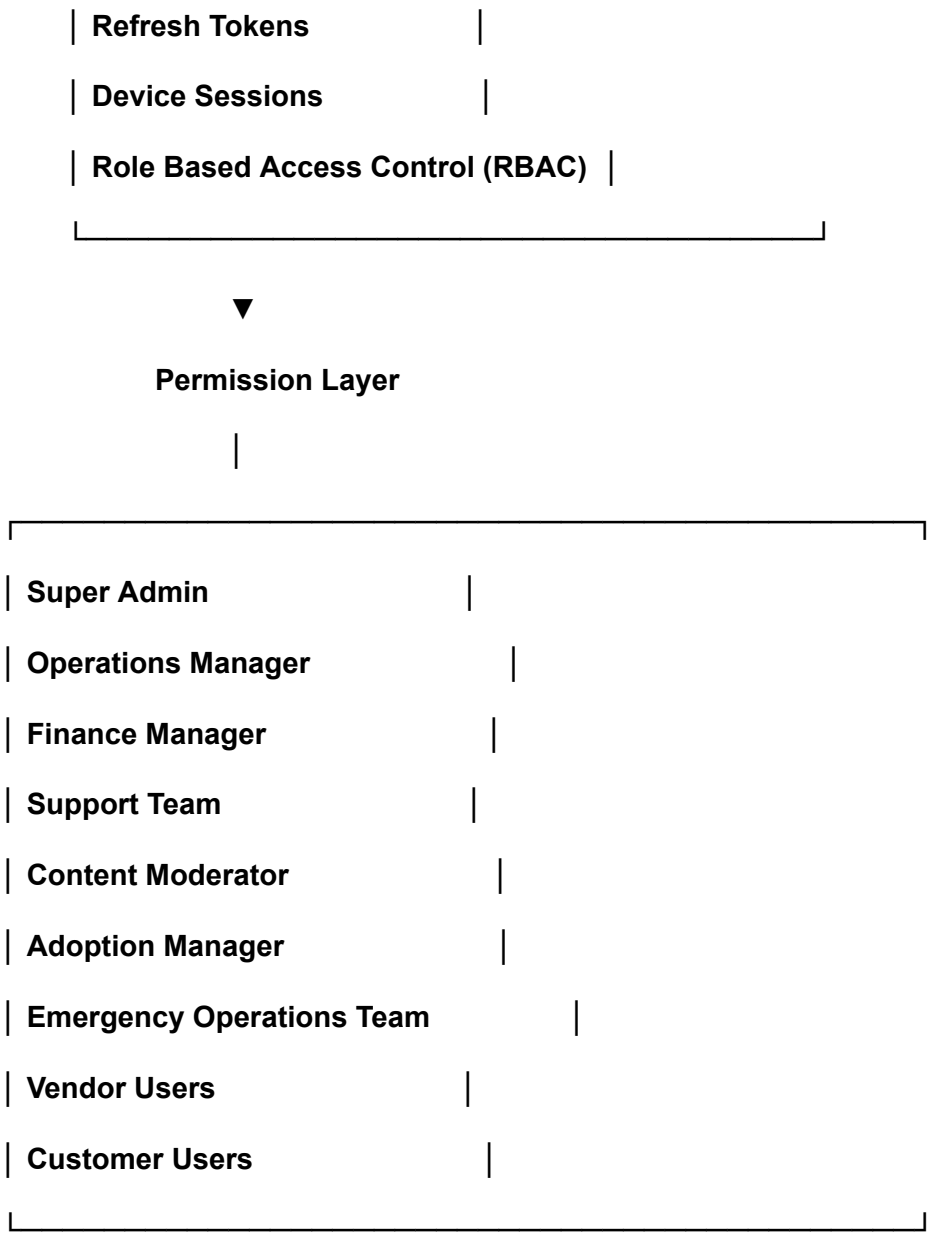
Rewards & Referral Architecture





Security & Access Architecture





Future Expansion Architecture

The architecture is intentionally designed for future migration toward:

- **AWS Cloud Infrastructure**
- **Kubernetes Clusters**
- **Microservices Architecture**
- **AI Recommendation Engine**
- **AI Image Matching**
- **Live Streaming**

- **Creator Monetization**
- **Subscription Memberships**
- **Geo-Fencing**
- **Multi-Country Deployment**
- **Multi-Language Support**
- **Franchise Operations**
- **Enterprise Analytics**

This ensures that PetSewa can scale from a city-level launch to a national pet ecosystem platform without major architectural changes.

Thank you for your consideration.

Regards,

Pearl Organisation